

Beginner Documentation Guide: MLB ParkCast

Project Overview

This repository contains all of the code and data required to run MLB ParkCast, an interactive dashboard of comprehensive performance analysis for all 30 Major League Baseball stadiums. This dashboard contains two tabs: analysis and projections. The projections tab allows users to input game-day details, like weather and start time, and then generates a daily park factor. The analysis tab contains detailed information on the stadium's physical characteristics and historical data on weather and play interactions. This section contains heat maps that show how baseball statistics change with weather, and a written analysis of weather trends at that stadium.

Repository Structure

The github workspace is divided into 4 main components: data cleaning, analysis, predictive modeling, dashboard design.

- The [data/](#) folder contains CSVs for each team-specific datasets of game-level statistics and weather data. These datasets are merged in `master_data.csv`.
- The [analysis/](#) folder contains the code that generates the weather heatmaps, weather distribution graphics, and follow-up analysis referenced in the stadium-specific write-ups.
- The predictive model lives in a unique branch of the github called [origin/model](#). This branch contains all of the code to run the predictive model for daily park factor and draws on the dataset from [data/](#) and introductory analysis in [analysis/](#).
- The dashboard itself can be opened in a browser via the [parkcast_dashboard.html](#) file in the origin branch of the github.

The analysis and model code are written in python. See [README.md](#) for required packages.

How to Use MLB ParkCast Github

1. Open the [README.md](#) file for the full repository map and technical appendix.
2. Open the dashboard – double click on [parkcast_dashboard.html](#) in any web browser.
3. Explore the data by looking at [master_data.csv](#) in [data/](#).
4. Review team-specific analysis in [analysis/](#). The [uniform_analysis.ipynb](#) file explains how to generate team specific graphics included in the dashboard.

Glossary

- Park Factor: A statistic (from Baseball Savant) measuring how much more or less offense a stadium produces versus a theoretical league-average park, where 100 = average. A value of 105 means 5% more run scoring than average.
- Daily/Weather-Adjusted Park Factor: A park factor predicted for a specific game given that day's expected conditions, rather than an annual aggregate.
- Day vs Night: Night games start after 4:00pm local time.
- Open-Air vs Covered: Covered stadiums include domes and retractable roofs; open-air stadiums have no roof structures.
- Marine Layer: Cool, humid, foggy air that rolls in off the Pacific (especially summer nights) at coastal parks. It produces denser air and stronger winds that suppress fly-ball carry.
- Away-Team Runs: This stat only tracks scores by the visiting team in order to avoid the roster bias of the home team and generate a more neutral measure of offense at the park.
- Pybaseball Statcast & MLB Stats API: Sources for pitch-level data, aggregated to game level.
- game_pk: The game ID used to aggregate pitch-level data into per-game rows.
- OpenMeteo: Source for hourly weather data pulled via a spatial grid around each stadium (3 hours post-first-pitch).
- Fixed Effects vs. Random Effects: Fixed effects capture average weather effects across all parks; random/park-dummy effects absorb baseline stadium differences.